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520

BASIC ELECTRONICS

ISCED UNIT CODE: 0714 441 04A
TVET CDACC UNIT CODE: IT/CU/ICTA/CC/01/5/MA

CONTINUOUS ASSESSMENT TEST (CAT 2)

78/100

TIME: 2 HOURS
TOTAL MARKS: 70

INSTRUCTIONS TO THE CANDIDATE

i. Answer all questions.

ii. Write your answers clearly and legibly.

iii. Draw well-labelled diagrams where required and show all working for calculations.

1. A technician is examining electronic components for use in a control circuit.

a) Define semiconductor material. (2 marks)

b) Describe the structure of a silicon atom, indicating the number of valence electrons. (3 marks)

c) Distinguish between conductors and semiconductors based on the energy band. (4 marks)

2. A trainee is preparing semiconductor materials for the manufacture of electronic components.

a) Describe how doping produces P-type semiconductor material. (3 marks)

b) Describe how doping produces N-type semiconductor material. (3 marks)

c) State two differences between P-type and N-type semiconductor materials. (4 marks)

3. A technician connects a PN-junction diode to a DC power supply to determine its operating characteristics.

a) Draw a labelled diagram of a PN-junction diode under forward bias. (4 marks)

b) Explain how forward bias affects the depletion region and current flow in the diode. (4 mark

c) Draw a labelled diagram of a PN-junction diode under reverse bias. (4 marks)

d) Explain how reverse bias affects the depletion region and current flow in the diode. (4 mar

4. Using the theorems of Boolean algebra , prove the following identities: (6marks)

(i) $(A+\overline{B})\cdot C+(B+\overline{C})\cdot A+\overline{A}\cdot B=A+B+C$

(ii) $\overline{X}\cdot(X+Z)+\overline{Y}+Y\cdot Z=\overline{Y}+Z$

5. Simplify the following functions of Boolean algebra using the k-map. (6marks)

(i) $F(a,b,c,d) = \sum (0,1,2,8,9,10)$

(ii) $F(a,b,c,d) = \sum (0,1,4,6,9,13,14,15)$

6. A computer system uses different storage technologies to store data and instructions.

a) Classify each of the following as semiconductor, magnetic or optical memory:

i. Solid-state drive (SSD) (1 mark)

ii. Hard disk drive (HDD) (1 mark)

iii. DVD (1 mark)

b) Distinguish between internal storage and online storage. (2 marks)

c) State two differences between main memory and secondary storage. (2 marks)

7. An engineer is designing a digital control circuit that receives two inputs, A and B.

a) Draw the symbols for the AND, OR and NOT gates. (3 marks)

b) Draw the truth table for an XOR gate. (3 marks)

c) State one difference between NAND and NOR gates. (2 marks)

d) Simplify the following Boolean expression using Boolean algebra. (4 marks)

$$Y = (A \cdot B) + (A \cdot \overline{B})$$

55/70

(1a) is a material whose electrical conductivity lies between that of a conductor and an insulator. Eg. silicon

(1b) silicon has an atomic number of 14 meaning it has 14 protons and 14 electrons in a neutral atom.
> its electrons are arranged in shells as 2,8,4.
> it has 4 valence electrons in its outermost shell

(1c) conductors: the valence and conduction bands overlap, allowing electrons to move freely ~~in neutral atom~~. while semiconductors: they have a small energy gap between the valence band and conduction band allowing electrons to conduct when they gain enough energy.

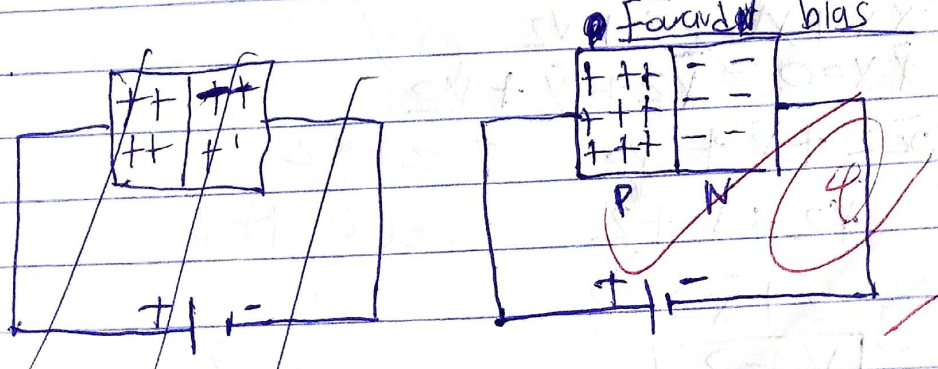
(2a) N-type material is formed by adding a pentavalent impurity such as phosphorous to pure silicon. This provides extra free electrons, which become the majority charge carriers.

(2b) P-type material is formed by adding a trivalent impurity, such as boron, to pure silicon. This creates holes that act as the majority charge carriers.

(2c)

P-Type	N-Type
* Holes are majority carriers	* Electrons are majority carrie
* Doped with trivalent impurities	* Doped with pentavalent impu

(3a)

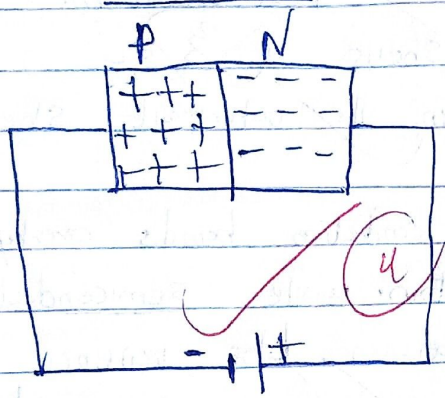


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(3b) The depletion region becomes narrower
> potential barrier decreases
> Majority charge carriers cross the junction more easily.
> Large forward current flows ones the diode's forward voltage is sufficient.

Reverse bias

(3c)



(3d) The depletion region beomes wider
> potential barrier increases
> majority carries are pulled away from the junction
> only a very small reverse leakage current flows until breakdown occurs.

(4i) $(\bar{A}+\bar{B})\cdot C + (B+\bar{C})\cdot A + \bar{A}\cdot B = A+B+C$

$= \bar{A}C + \bar{B}C + AB + A\bar{C} + \bar{A}B$

$= \bar{A}C + \bar{B}C + B + A\bar{C}$

$= \bar{A}C + B + A\bar{C}$

using $\bar{A}C + A\bar{C} = \bar{A}C + \bar{A}C$ and absorption:

$= \boxed{A+B+C}$

(4ii) $\bar{x}(x+z) + \bar{y} + xz = \bar{y}+z$

$= \bar{x}x + \bar{x}z + \bar{y} + yz$

$= \bar{x}x = 0 \quad = \bar{x}z + \bar{y} + yz$

$= \bar{x}z + \bar{y} + yz$

$= \bar{x}z + \bar{y} + z$

using $z + \bar{x}z = z$

$= \boxed{\bar{y}+z}$

$\bar{x}z + xz$

$z(\bar{x}+x) + \bar{y}$

1

$z + \bar{y}$

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5iv $F(A,B,C,D) = \Sigma(0,1,2,8,9,10)$

AB \ CD	00	01	11	10
00	1	1	0	1
01	0	0	0	0
11	0	0	0	0
10	1	1	0	1

$\therefore F = \bar{B}\bar{C} + \bar{B}\bar{D}$

i) $F = \bar{B}(\bar{C} + \bar{D})$

(5ii) $F(abcd) = \Sigma m(0,1,4,6,9,13,14,15)$

AB \ CD	00	01	11	10
00	1	1	0	0
01	1	0	0	1
11	0	1	1	1
10	0	1	0	0

ii) $F = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{D} + A\bar{C}D + ABC$

(6a) (i) SSD - semi conductor
(ii) DVD - optical
(iii) HDD - Magnetic

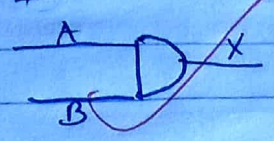
(6b) Internal storage is data stored in a physical drive e.g HDD while online storage is data stored in the internet / remotley and can be accessed through internet e.g google drive.

(6c Main memory
> usually faster to access
> Ram is volatile - data is lost when power is off.

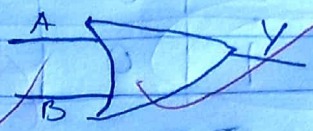
> usually retains

Secondary memory
slower to access
data without power.

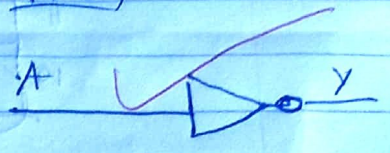
(7a AND



OR



NOT



(7b Truth table XOR

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(7c NAND - produces 0 only when all inputs are 1
> NOR - produces 1 only when all inputs are 0

(7d $Y = (A \cdot \bar{B}) + (\bar{A} \cdot B)$

$Y = A(\bar{B} + B)$

$Y = A \cdot 1$

$\boxed{Y = A}$